DISCRETE MATH · THE LOGIC OF COMPOUND STATEMENTS

Truth Table Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Truth Table Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. How many rows does a truth table need for a statement with 3 variables? With 4?
   8 rows for 3 variables (2^3); 16 rows for 4 (2^4).
2. Predict: is "if p then q" equivalent to "if q then p"? Explain in a sentence.
   No. The converse can be false when the original is true (for example p false, q true makes p → q true but q → p false).

2 Explore use the tool

1. Statement 1: p → q. Use the buttons to make the converse, inverse and contrapositive. Which are equivalent to p → q?
   Only the contrapositive ~q → ~p. The converse and inverse are not (they differ in the rows p = T, q = F and p = F, q = T).
2. Compare the converse and the inverse with each other (type q → p and ~p → ~q). What do you find?
   They are equivalent to each other (the inverse is the contrapositive of the converse).
3. Type ~(p → q) and find an equivalent statement with no arrow. Record it.
   p ∧ ~q.
4. Test an argument mode: p ∨ q; ~p, so q. How many critical rows are there, and is it valid?
   One critical row (p = F, q = T), where the conclusion is true, so it is valid (disjunctive syllogism).

3 Record use the tool and fill in the table

Classify each statement

Use Statement 1 only. Write tautology, contradiction or contingency, and the number of true rows.

Statement	Rows	True rows	Type
p ∨ ~p	2	2	tautology
p ∧ ~p	2	0	contradiction
(p → q) ∨ (q → p)	4	4	tautology
p ↔ q	4	2	contingency
(p ∧ q) → r	8	7	contingency

Valid or invalid?

Premises	Conclusion	Critical rows	Valid?
$p \to q$; p	q	1	valid (modus ponens)
$p \to q$; q	p	2	invalid (converse error)
$p \to q$; $\sim p$	$\sim q$	2	invalid (inverse error)
$p \to q$; $q \to r$	$p \to r$	4	valid (transitivity)

4 Explain what did you discover?

1. Two statements are logically equivalent when their columns agree in **every** row.
2. $p \to q$ is equivalent to its **contrapositive**, but not to its converse or its **inverse**.
3. $\sim(p \to q)$ is equivalent to **$p \wedge \sim q$**.
4. An argument form is valid when the conclusion is **true** in every critical row (every row where all the premises are **true**).
5. A classmate says, "My argument has a true conclusion in one critical row, so it is valid." Explain the mistake.
   Validity needs the conclusion true in every critical row. One critical row with a false conclusion makes the form invalid, no matter how many other rows work.

5 Apply a new situation, no tool

1. Write the contrapositive of "If a number is divisible by 6, then it is even." Is it true?
   If a number is not even, then it is not divisible by 6. True (it is equivalent to the original, which is true).
2. Write the negation of "If it rains, then the game is canceled" without using "if".
   It rains and the game is not canceled.
3. Is $(p \to q) \wedge (q \to r) \to (p \to r)$ a tautology? How many rows do you need to check?
   Yes, it is a tautology; 8 rows (3 variables).
4. "If I study, I pass. I passed. So I studied." Name the form and say whether it is valid.
   Converse error (affirming the consequent): invalid. In the row study = F, pass = T both premises are true but the conclusion is false.

Teacher notes

Intended discovery: Equivalence means identical truth-table columns; the contrapositive is equivalent to the conditional while the converse and inverse are equivalent to each other; $\sim(p \to q) \equiv p \wedge \sim q$; validity is about critical rows.

Likely misconceptions:

- Believing the converse of a true conditional is true.
- Negating $p \to q$ as $\sim p \to \sim q$ (that is the inverse).
- Judging validity by whether the conclusion sounds true instead of by the critical rows.

Answers that can vary: Explore 3 can be any statement equivalent to $p \wedge \sim q$ (for example $\sim(\sim p \vee q)$). Explain wording varies.